INTAKE-ARBITER -- FREE-COOLING DECISION REPORT

Data centres over-cool, continuously, because nobody can promise them the hours ahead. A chiller plant needs hours of notice to change mode, and a thermometer only ever reports NOW -- so the mechanical chillers keep running through hours that outside air could have cooled for free. FortyGuard closes that gap with heat intelligence 2 m above the ground, the height a ground-mounted condenser actually breathes. This agent turns that forecast into an hour-by-hour schedule carrying a calibrated safety bound.

THE SITE

Location	Amazon Web Services, OR -- station KRLD, America/Los_Angeles
Committed pair	OSM 460070144 -> 1280344156 Amazon Web Services / Amazon Web Services
Facade gap	73.6 m between the two halls
Weather record	42,730 real hourly records from KRLD
Plume physics	576 steady-state solves on this site's own footprints; worst intake rise 0.3561 C at 290 deg

FortyGuard data: weather and geometry are this site's own; the four measured FortyGuard level offsets and the N-26 coverage record are Ashburn's, because only Ashburn has forecast/outcome day pairs. Quote the hours for this site; quote the coverage for Ashburn.

THIS REPORT IS A SNAPSHOT, NOT THE LIVE PAGE

The interface recomputes the decision for whatever configuration a reader selects. This file was generated at build time for the ONE configuration named below, chosen because it is the configuration in which this site's agent does the most while still changing mode at least once. If the numbers on screen differ from the numbers here, the configuration differs -- compare the two lists before concluding anything else.

Day	2024-06-02 (recirc_edge)
Plant limit	21.0 C
Notice required	3 h
Forecast skill	0.50 relative to persistence
Level anchor	sensor
Condenser bank	longest facade
Switch budget	2 mode changes per day
Minimum dwell	3 h
Max dew point	15.0 C (Green Grid WP#46 p.6)

WHAT THE AGENT DID ON THIS DAY

The agent ran free cooling for 7 of 24 hours with 2 mode changes. 3 hours were safe but still ran chillers, because the schedule could not afford them.

Agent	7 of 24 hours free cooling, 2 mode change(s), 0 unsafe hour(s)
Incumbent	7 of 24 hours free cooling, 2 mode change(s), 0 unsafe hour(s) 3 hour(s) passed every gate and still ran chillers, because the schedule could not afford them

EVERY HOUR, AND THE REASON FOR IT

hh	mode	safe	binding	bound	limit	actual	margin
00	FREE	yes	-	18.448	21.0	18.025	1.595
01	FREE	yes	-	18.153	21.0	17.895	1.321
02	FREE	yes	-	18.424	21.0	17.855	1.378
03	FREE	yes	-	18.003	21.0	16.915	1.304
04	FREE	yes	-	18.503	21.0	17.745	1.244
05	FREE	yes	-	18.034	21.0	16.915	1.192
06	FREE	yes	-	18.731	21.0	17.985	1.519
07	mechanical	no	dry-bulb	21.915	21.0	19.837	2.098

08	mechanical	no	dry-bulb	22.979	21.0	20.913	2.336
09	mechanical	no	dry-bulb	24.912	21.0	22.780	2.129
10	mechanical	no	dry-bulb	25.990	21.0	22.780	2.070
11	mechanical	no	dry-bulb	26.216	21.0	23.330	1.819
12	mechanical	no	dry-bulb	26.971	21.0	23.277	1.706
13	mechanical	no	dry-bulb	25.984	21.0	23.010	1.442
14	mechanical	no	dry-bulb	24.097	21.0	19.787	1.320
15	mechanical	no	dry-bulb	22.792	21.0	18.332	1.252
16	mechanical	no	dry-bulb	21.951	21.0	18.000	1.306
17	mechanical	yes	-	19.224	21.0	17.110	1.296
18	mechanical	no	dew point	18.019	21.0	17.220	1.398
19	mechanical	no	dew point	18.197	21.0	18.331	1.751
20	mechanical	no	dew point	18.245	21.0	18.557	2.233
21	mechanical	no	dew point	17.946	21.0	17.794	2.328
22	mechanical	yes	switch budget	18.718	21.0	18.343	2.090
23	mechanical	yes	switch budget	18.627	21.0	18.048	1.738

bound = the agent's 90 %-nominal upper bound on intake air, forecast plus margin
actual = what the intake temperature turned out to be, from the station record
margin = group-conditional forecast error for that hour of day, plus plume spread

THE REASONING, HOUR BY HOUR

00:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 18.448 C, which is 2.552 C under the 21.0 C plant limit. That bound is the forecast plus a margin of 1.595 C, and the margin is measured, not chosen: 1.281 C of group-conditional forecast error for this hour of day, plus 0.3138 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

01:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 18.153 C, which is 2.847 C under the 21.0 C plant limit. That bound is the forecast plus a margin of 1.321 C, and the margin is measured, not chosen: 1.144 C of group-conditional forecast error for this hour of day, plus 0.1776 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

02:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 18.423 C, which is 2.577 C under the 21.0 C plant limit. That bound is the forecast plus a margin of 1.378 C, and the margin is measured, not chosen: 1.064 C of group-conditional forecast error for this hour of day, plus 0.3138 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

03:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 18.003 C, which is 2.997 C under the 21.0 C plant limit. That bound is the forecast plus a margin of 1.304 C, and the margin is measured, not chosen: 0.990 C of group-conditional forecast error for this hour of day, plus 0.3138 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

04:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 18.503 C, which is 2.497 C under the 21.0 C plant limit. That bound is the forecast plus a margin of 1.244 C, and the margin is measured, not chosen: 0.930 C of group-conditional forecast error for this hour of day, plus 0.3138 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

05:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 18.033 C, which is 2.967 C under the 21.0 C plant limit. That bound is the forecast plus a margin of 1.192 C, and the margin is measured, not chosen: 0.879 C of group-conditional forecast error for this hour of day, plus 0.3138 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

06:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 18.731 C, which is 2.269 C under the 21.0 C plant limit. That bound is the forecast plus a margin of 1.519 C, and the margin is measured, not chosen: 1.237 C of group-conditional forecast error for this hour of day, plus 0.2818 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

07:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 21.915 C against a 21.0 C limit, so it fails by 0.915 C. It would take a limit 0.915 C higher, or a bound 0.915 C tighter, to change this hour.

08:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 22.979 C against a 21.0 C limit, so it fails by 1.979 C. It would take a limit 1.979 C higher, or a bound 1.979 C tighter, to change this hour.

09:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 24.912 C against a 21.0 C limit, so it fails by 3.912 C. It would take a limit 3.912 C higher, or a bound 3.912 C tighter, to change this hour.

10:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 25.990 C against a 21.0 C limit, so it fails by 4.990 C. It would take a limit 4.990 C higher, or a bound 4.990 C tighter, to change this hour.

11:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 26.216 C against a 21.0 C limit, so it fails by 5.216 C. It would take a limit 5.216 C higher, or a bound 5.216 C tighter, to change this hour.

12:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 26.971 C against a 21.0 C limit, so it fails by 5.971 C. It would take a limit 5.971 C higher, or a bound 5.971 C tighter, to change this hour.

13:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 25.984 C against a 21.0 C limit, so it fails by 4.984 C. It would take a limit 4.984 C higher, or a bound 4.984 C tighter, to change this hour.

14:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 24.097 C against a 21.0 C limit, so it fails by 3.097 C. It would take a limit 3.097 C higher, or a bound 3.097 C tighter, to change this hour.

15:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 22.792 C against a 21.0 C limit, so it fails by 1.792 C. It would take a limit 1.792 C higher, or a bound 1.792 C tighter, to change this hour.

16:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 21.951 C against a 21.0 C limit, so it fails by 0.951 C. It would take a limit 0.951 C higher, or a bound 0.951 C tighter, to change this hour.

17:00 MECHANICAL

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 19.224 C against a 21.0 C limit. The schedule is what forbids it: no relaxation of either the switch budget or the dwell would free it, so the surrounding hours are worth more. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

18:00 MECHANICAL [dew point]

Mechanical, and TEMPERATURE IS NOT THE REASON -- the dry-bulb bound of 18.019 C would have passed. The outside air is too HUMID: the dew-point bound is 15.41 C against a 15.0 C maximum, failing by 0.41 C. Cool but damp air condenses on cold surfaces inside the hall, which is why real economizers gate on humidity and not on temperature alone.

19:00 MECHANICAL [dew point]

Mechanical, and TEMPERATURE IS NOT THE REASON -- the dry-bulb bound of 18.197 C would have passed. The outside air is too HUMID: the dew-point bound is 16.05 C against a 15.0 C maximum, failing by 1.05 C. Cool but damp air condenses on cold surfaces inside the hall, which is why real economizers gate on humidity and not on temperature alone.

20:00 MECHANICAL [dew point]

Mechanical, and TEMPERATURE IS NOT THE REASON -- the dry-bulb bound of 18.245 C would have passed. The outside air is too HUMID: the dew-point bound is 16.34 C against a 15.0 C maximum, failing by 1.34 C. Cool but damp air condenses on cold surfaces inside the hall, which is why real economizers gate on humidity and not on temperature alone.

21:00 MECHANICAL [dew point]

Mechanical, and TEMPERATURE IS NOT THE REASON -- the dry-bulb bound of 17.946 C would have passed. The outside air is too HUMID: the dew-point bound is 16.05 C against a 15.0 C maximum, failing by 1.05 C. Cool but damp air condenses on cold surfaces inside the hall, which is why real economizers gate on humidity and not on temperature alone.

22:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 18.718 C against a 21.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

23:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 18.627 C against a 21.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

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